Zhanggir Yessenaliyev

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Education | Available Spring and Summer 2027

Purdue University | West Lafayette, IN
B.S. in Robotics Engineering Technology | *Aug 2025 – May 2029*

Technical Skills

Languages: C++17, C, Python, MATLAB, Bash, HTML/CSS
AI/Computer Vision: PyTorch, CUDA, OpenCV, NumPy, Ultralytics YOLO, Image Segmentation
Robotics: ROS 2, CARLA, Nav2, Gazebo, SLAM Toolbox, TF2, Localization, SLAM, Sensor Fusion, LiDAR
Developer Tools: Linux, Git, Docker, CMake, RViz2, Foxglove
CAD/CAE: SOLIDWORKS, NX, Autodesk Fusion 360

Experience

Purdue IEEE Autonomous Kart Racing | Jan 2026 – Present
Controls & Perception Team Member | *West Lafayette, IN*

- Built a 30 Hz ROS 2 localization stack fusing ZED VIO and IMU via an EKF with 100 ms sensor timeouts.
- Migrated odom to base_link TF ownership from the ZED node to robot_localization for a robot-centric tree.
- Configured ZED GEN_3 tracking with gravity-aligned origin and 2D mode, enabling autonomous kart testing.
- Enabled deployment of the kart at the Autonomous Kart Series by validating its localization systems.

Autonomous Robotics Club of Purdue | Sep 2026 – Present
Mechanical Electrical Engineer | *West Lafayette, IN*

- 3D-printed and assembled a gimbal mount for a ZED camera; currently designing a custom controller PCB.
- Designed an enclosure for the NVIDIA Jetson Orin Nano in CAD.

SilkRoad Innovation Hub – Early-Stage VC | Jul 2026 – Aug 2026
Robotics Fellow | *Menlo Park, CA*

- Earned selection as 1 of 10 Fellows from a competitive pool of 120+ applicants.
- Presented robotics projects to engineers from Uzbekistan's Ministry of Education and advised them on investor readiness.

Engineering Club | Oct 2024 – Jun 2025
Science Mentor | *Almaty, Kazakhstan*

- Recruited and led a 15-student engineering team, assigning development responsibilities for a biomechanical hand.
- Designed, prototyped, and tested an EMG-controlled biomechanical hand using an Arduino-based control system.
- Placed 3rd among 656 projects at InfoMatrix Asia.

Projects

Vision-Guided Autonomous Navigation | *Python, PyTorch, CUDA, CARLA, OpenCV* | Jul 2026 – Sep 2026

- Built an autopilot-free, route-aware controller that achieved 98.9% mean completion across 9 weather-varied runs over 1.48 km with 0 collisions and 0 solid-line crossings.
- Accelerated model inference 20.4x over CPU to 199 FPS with 4.72 ms median latency using CUDA mixed precision.
- Trained a 117K-parameter U-Net-style deep neural network in PyTorch on 900 synchronized CARLA RGB/semantic pairs, achieving 92.3% IoU and 96.0% Dice on 300 held-out images.

Occlusion-Aware Augmented Reality Localization | *Python, OpenCV, NumPy, ArUco* | Jul 2026 – Present

- Built a world-space camera-geometry pipeline implementing lens correction, coordinate transforms, and monocular ground-plane projection.
- Achieved 0.57-pixel calibration RMS across 20+ views and implemented ArUco/solvePnP camera localization with reprojection-error diagnostics.

Nav2 Custom Controller | *C++, ROS 2 Jazzy, Nav2, pluginlib, TF2* | Jun 2026 – Aug 2026

- Implemented a 20 Hz C++ Pure Pursuit path-tracking algorithm using Nav2 global paths and robot poses to generate bounded velocity commands in 0.2 ms per control cycle.
- Integrated pluginlib lifecycle management, TF2 transformations, costmap-aware path pruning, RViz visualization, and 10 YAML-configurable motion parameters.
- Automated TurtleBot3 testing across 20 navigation goals by scripting simulation launch, AMCL initialization, navigation-goal dispatch, and headless or graphical execution.

Autonomous Differential Rover Sim | *ROS 2 Jazzy, Nav2, Gazebo, SLAM, URDF/Xacro, Python* | Jul 2026

- Built a URDF/Xacro differential-drive rover with Gazebo physics, 50 Hz odometry, and 360-degree LiDAR.
- Bridged six Gazebo–ROS 2 channels for control, odometry, TF, joint states, LiDAR, and simulation time.
- Configured SLAM Toolbox for 5 cm occupancy-grid mapping with scan matching and loop closure.

Little Blazer Engine CAD Assembly | *Siemens NX, Teamcenter, PDM/PLM* | May 2026

- Built a complete Little Blazer Engine assembly in Siemens NX using parametric parts and subassemblies.
- Produced engineering drawings and 3D annotations documenting component geometry and assembly relationships.
- Managed revision-controlled parts and assembly data in Teamcenter using PDM/PLM workflows.